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Q. Find the area common to the two parabola $x^2 = 4ay$ and $y^2 = 4ax$

Solution: Given that,

$$x^2 = 4ay \rightarrow \textcircled{1}$$

$$y^2 = 4ax \rightarrow \textcircled{2}$$

Now from $\textcircled{1}$ & $\textcircled{2}$ we have,

$$x^2 = 4a \cdot \sqrt{4ax}$$

$$\text{or, } x^4 = 16a^2 \cdot 4ax$$

$$\text{or, } x^4 - 64a^3x = 0$$

$$\text{or, } x(x^3 - 64a^3) = 0$$

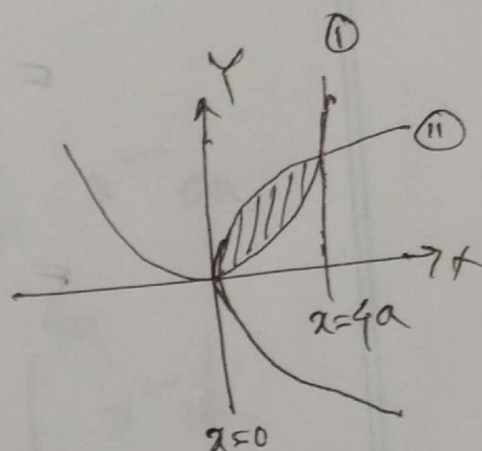
$$\text{or, } x = 0, x^3 - 64a^3 = 0$$

$$\text{or, } x^3 = (4a)^3$$

$$\text{or, } x = 4a$$

$$\therefore x = 0, x = 4a$$

Hence the required area, $A = \int_{x=0}^{4a} (y_1 - y_2) dx$



$$= \int_{x=0}^{4a} \left(\sqrt{4ax} - \frac{x^2}{4a} \right) dx$$

$$= \int_0^{4a} 2\sqrt{a} \cdot \sqrt{x} \, dx - \int_0^{4a} \frac{1}{4a} \cdot x^2 \, dx$$

$$= 2\sqrt{a} \int_0^{4a} x^{\frac{1}{2}} \, dx - \frac{1}{4a} \int_0^{4a} x^2 \, dx$$

$$= 2\sqrt{a} \cdot \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^{4a} - \frac{1}{4a} \left[\frac{x^3}{3} \right]_0^{4a}$$

$$= 2\sqrt{a} \cdot \frac{2}{3} [(4a)^{3/2} - 0] - \frac{1}{4a} \cdot \frac{1}{3} [(4a)^3 - 0]$$

$$= \frac{4\sqrt{a}}{3} \cdot 8a\sqrt{a} - \frac{1}{12a} \cdot 64a^3$$

$$= \frac{32a^2}{3} - \frac{16a^2}{3}$$

$$= \frac{16a^2}{3}$$

Ans.